Ogbonna Arinze

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SUMMARY

Senior Fullstack Engineer with 8 years of professional experience building secure, scalable web applications and embeddable components across healthcare, fintech, e-commerce, and civic tech environments. Deep expertise in **authentication and identity systems**, including JWT-based auth flows, role-based access control, session management with HTTP-only cookies, and OTP-based authentication. Proven track record of building cross-platform embeddable widgets and Chrome extensions in **React** and **TypeScript**, backed by production **Node.js** microservices, **RESTful APIs**, and **AWS** infrastructure. Passionate about ownership-driven engineering with end-to-end delivery from frontend to infrastructure.

CORE COMPETENCIES

- **Authentication & Security:** JWT authentication, role-based access control (RBAC), HTTP-only cookie session management, OTP authentication, secure token refresh flows, and permission-based UI rendering.
- **Frontend:** React, TypeScript, JavaScript (ESNext), Next.js, Vue.js, HTML, CSS, Tailwind CSS, Radix UI, and design systems.
- **Backend & APIs:** Node.js, NestJS, RESTful API design, microservice architecture, PostgreSQL, MongoDB, Prisma, and GraphQL.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, Docker Compose, GitHub Actions, CI/CD pipelines, Nx monorepo orchestration, and Traefik reverse proxy.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), and agentic workflows.
- **Leadership:** Technical ownership ("you build it, you own it"), architecture reviews, mentorship, Agile/Scrum, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer — *May 2023 - Present*

- Implemented **JWT-based authentication** and **role-based access control** on the frontend, securely handling access and refresh tokens with **HTTP-only cookies**, integrating login, signup, and permission-based UI rendering, and collaborating with backend engineers to align on auth strategies.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension** embedded across clinical site systems, delivering a cross-platform component integrated into external workflows.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, designed as an embeddable widget that users could integrate on external sites to query their data in real time.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria

Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**, developing critical **e-commerce**, MasterCard, VISA, and Express-cash payment modules.
- Developed the e-commerce payment module using **React** with **Redux** for state management, **RxJS** for real-time data streaming, and **Axios** for asynchronous API integration with backend microservices.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)

Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** to power new features, focusing on testable, maintainable, and well-documented code, and translated business requirements into efficient **MongoDB** database models and schemas.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria

Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.

PROJECTS

OurNigeria ournigeria.ng

Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with pnpm workspaces.
- Implemented **phone OTP authentication** with 6-digit codes and session cookies, **Telegram OAuth** login, and a **Telegram bot** with webhook-based message handling, demonstrating production identity and authentication system design.
- Built the **NestJS** backend with **RESTful API** design, **PostgreSQL 16** with **Prisma v7**, and **SSE streaming** for real-time chat responses with 15s keepalive heartbeats to prevent reverse proxy timeouts.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and **Traefik** for reverse proxy routing.
- Built a **multi-agent AI system** using **Mastra** with 5 specialized agents and a hybrid **RAG pipeline** combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*